



**IN
PARTNERSHIP
WITH
PLYMOUTH
UNIVERSITY**

**PUSL3122 HCI, Computer
graphics and visualisation**

Topics covered last Lecture (Lecture 5) ...

- **Prototyping**

- Conceptual design, Concrete design, Using scenarios, Generating prototypes, Storyboard, User experience maps, Customer journeys
- This part of the module is taken from the main text book for the module, the latest release from 2019. This lecture focuses on chapter 12.



Chapter 12
DESIGN, PROTOTYPING and
CONSTRUCTION

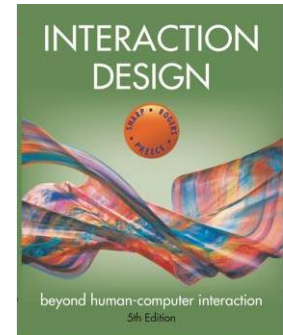
Topics covered this lecture...(Lecture 6)

- **The Practice of Interaction Design**

- Agile UX refers to approaches that **integrate** UX design and agile development including Design patterns, Open-source resources, Tools to support interaction design

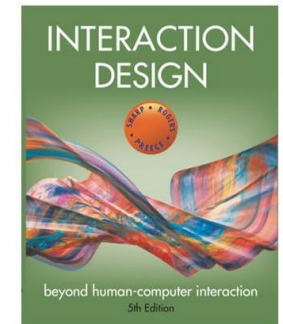
- **Discovering requirements**

- A requirement is a **statement** about an **intended product** that specifies **what it is expected to do** or **how it will perform**.
- Articulating requirements, Different kinds of requirements, Requirements data gathering, Scenarios, Personas.
- Scenarios and personas together **bring requirements to life**.
- **Use cases** capture details.



Chapter 13

INTERACTION
DESIGN
IN PRACTICE



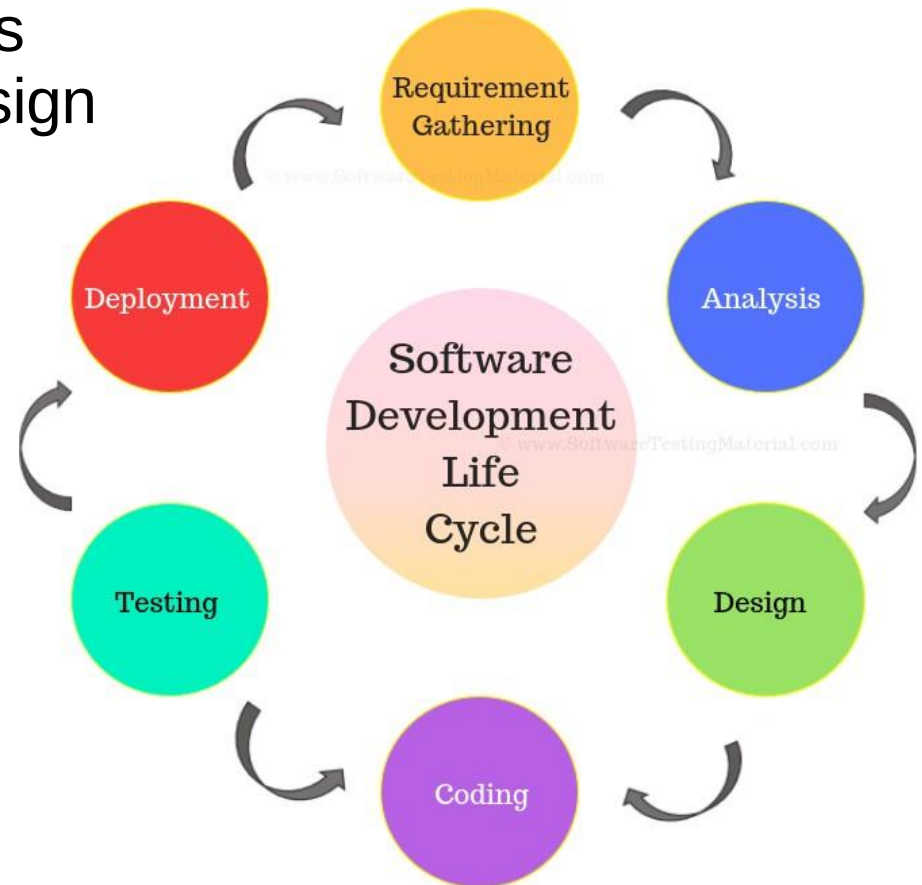
Chapter 11

DISCOVERING
REQUIREMENTS

The Practice of Interaction Design

Overview

1. Agile development
2. Design Patterns
3. Open Source Resources
4. Tools for Interaction Design



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1. Agile development

The Practice of Interaction Design

Overview

1. Agile development
2. Design Patterns
3. Open Source Resources
4. Tools for Interaction Design

- Short (one to three week) time boxes of iterative development (**sprint**, iteration, cycle)

- **Early** and **repeated** customer/user feedback



- **Re-prioritization of work** based on customer or user so that emergent requirements can be handled
- **Many approaches**, for example, eXtreme Programming (XP), Scrum, Kanban, and DSDM

Technical Debt in UX

- Making **compromises** that are **expedient in short term**, but cause **problems in long term**
- Refactoring to correct pragmatic trade-offs
- **UX debt** arises from trade-offs

- A history of neglecting UX



- Large portfolio of products developed independently

AgileUX

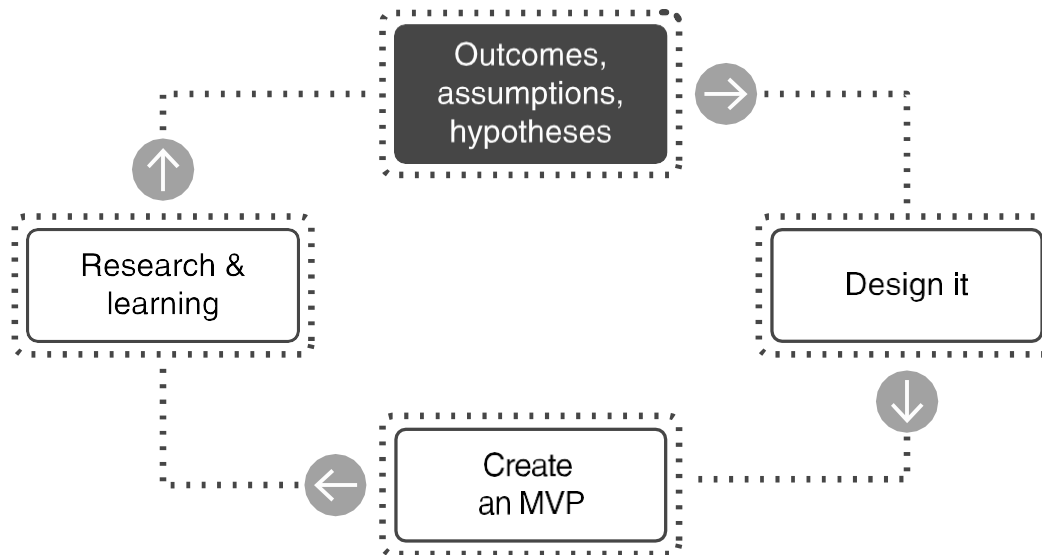
- Integrates techniques from **interaction design** and **agile** methods
- AgileUX requires **balancing** research and **reflection** for good UX with **rapid iterations** incorporating user feedback
- In Agile iterations, requirements are elaborated (**evolve**) and **re-prioritized**, rather than specified upfront
- All techniques in UX are still relevant, but when to use and how much needs careful planning
 - Focus on **product**, not **design**, as deliverable
 - **Cross-functional** teams 
- Three practical areas: user research, aligning work practices, and documentation

User research

- Seeks to **characterize users**, tasks, and context through **data gathering** and **analysis**
- Detailed user research **cannot be fitted** within a limited time box
 - Some user research **can be performed** in iteration 0 (zero), **before** implementation starts
 - Ongoing program of user research over a longer period of time

LeanUX

- **Create** and **deploy** products **quickly** to get real user feedback for future developments
- **Tight** iterations of build-measure-learn
- Minimum viable product **tests** assumptions

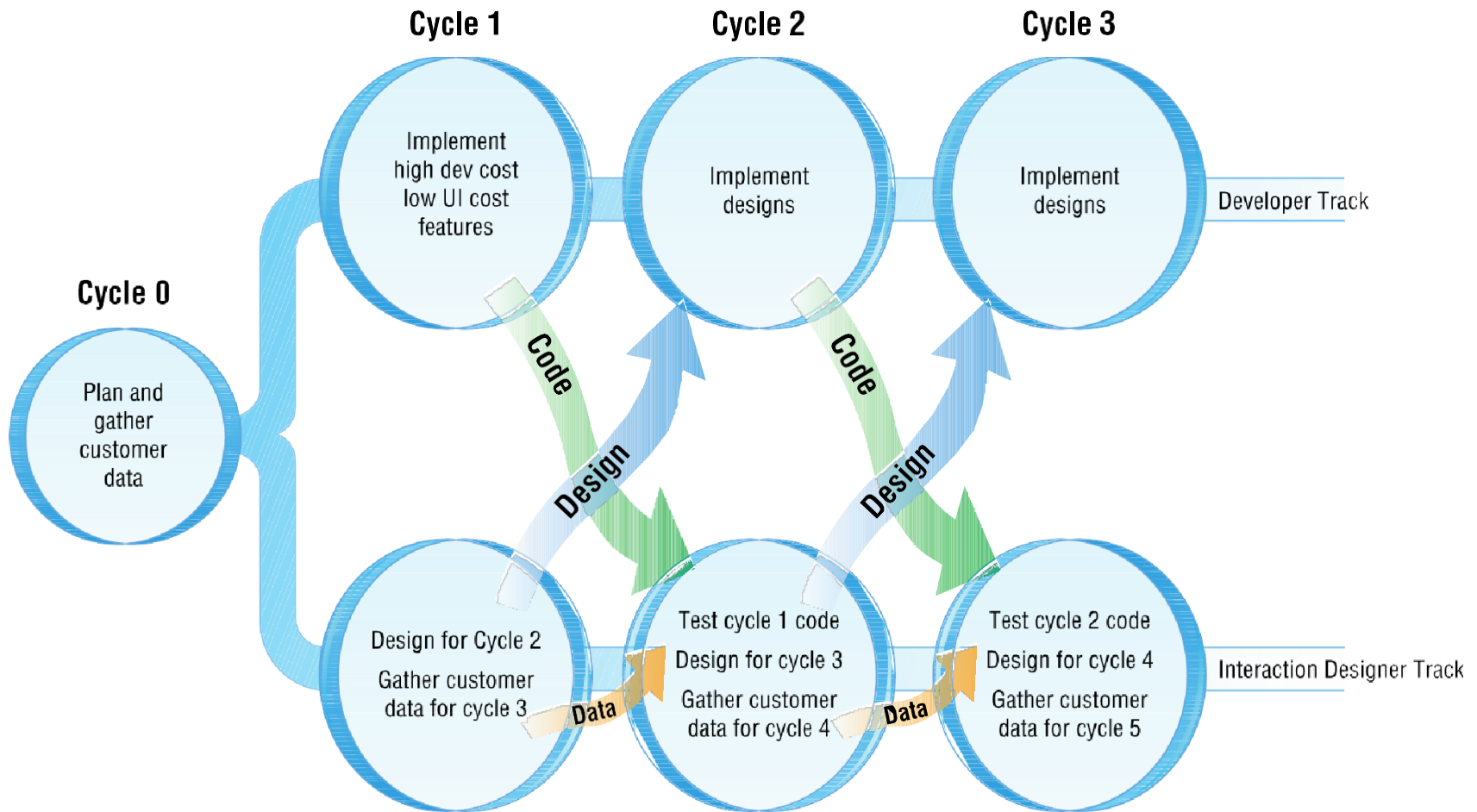


Source: Gothelf, J. with J. Seiden (2016) Lean UX, 2nd edition, O'Reilly.

Aligning work practices

- Designing a complete product upfront causes problems because of re-prioritization
- **Some** upfront work is needed (technical and UX)
- Use a parallel tracks approach:
 - Create **product vision** before development starts
 - Do **design** work **one iteration ahead** of development
 - Some teams work **two iterations** ahead

Parallel tracks approach to AgileUX



Source: Sy, D. (2007) Adapting usability investigations for development, *Journal of Usability Studies* 2(3), May, 112–130. [User Experience Professionals Association](#).

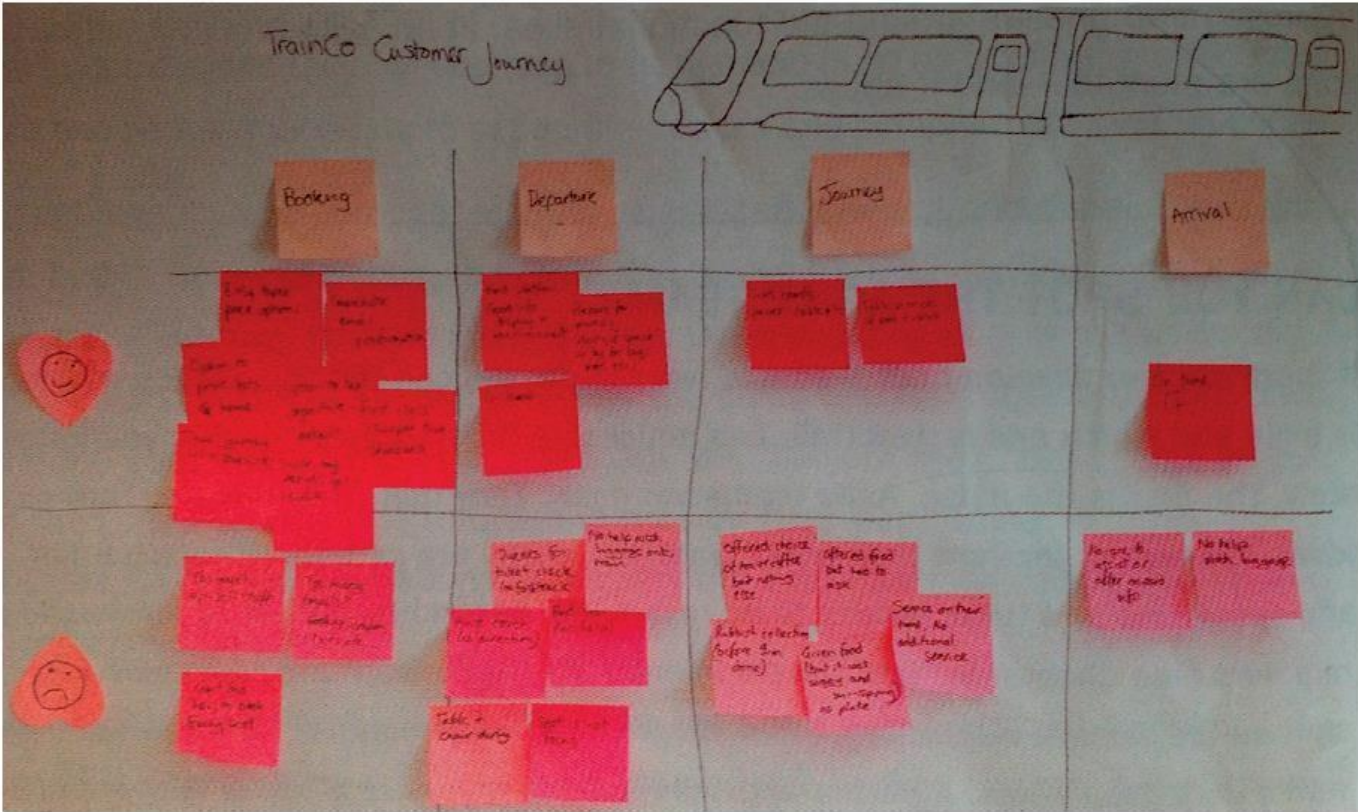
Aligning work practices

- Advantages of parallel tracks approach:
 - No design time wasted on features not implemented
 - **Usability testing** and **contextual inquiry** could be done on the same customer visit, saving time
 - Timely feedback on the designs was received from developers and customers
 - Agile flexibility supports schedule changes if a problem is found
- Parallel tracks is **commonly used**

Documentation

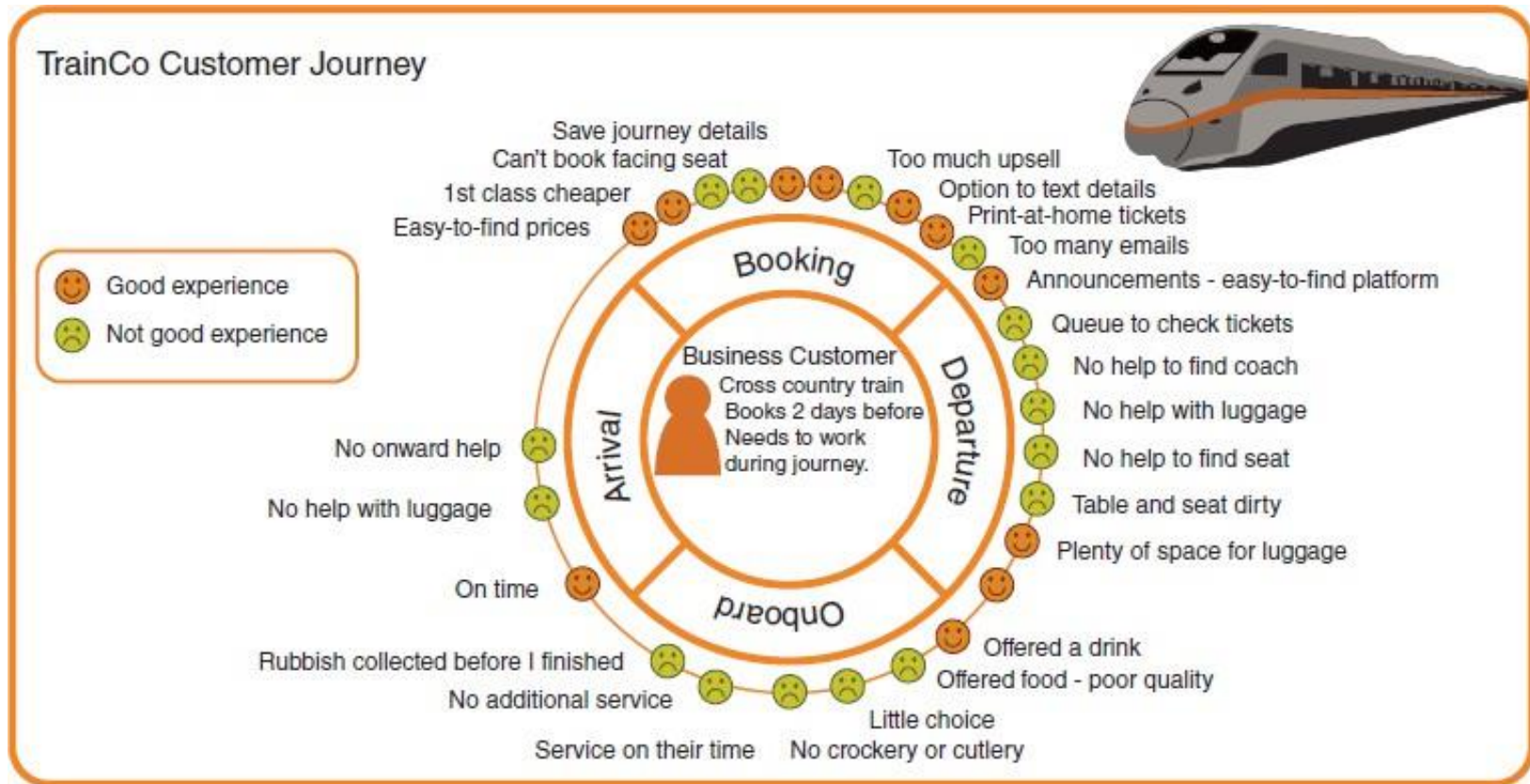
- Common **communication** approach for UX designers
- Agile encourages **minimal documentation** so more time can be spent on **design** and **discussion**
- Only use documentation where needed. Ask:
 - Can the amount of time spent on documentation be **reduced**?
 - Who uses documentation?
 - What is the **minimum** needed by customers?
 - How efficient is the sign-off process?
 - Is there duplication anywhere?
 - **How polished does documentation need to be?**

Documentation: How polished? (low fidelity)



Source: Ratcliffe, L. and McNeill, M. (2012) Agile Experience Design. New Riders.

Documentation: how polished? (high fidelity)



Source: Ratcliffe, L. and McNeill, M. (2012) Agile Experience Design. New Riders.

2. Design Patterns

The Practice of Interaction Design

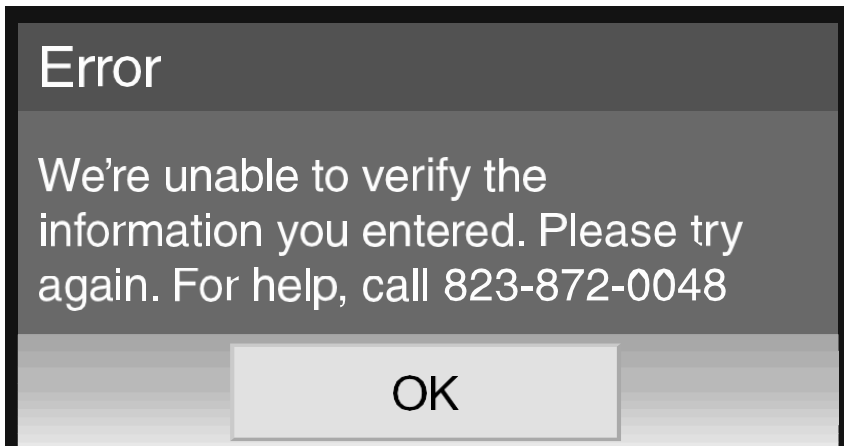
Overview

1. Agile development
2. Design Patterns
3. Open Source Resources
4. Tools for Interaction Design

- *(not software implementation like Observer etc.)*
- Capture design experience:
 - A solution to a problem in a context
 - Can be instantiated in many ways: generative
- Patterns may be individual, in languages, in catalogues, galleries, or libraries
- Patterns often are associated with software components, for example, GitHub or platform websites

Bad Design Patterns

- Capture design experience — but that doesn't necessarily mean good design:
 - Anti-patterns: Don't do it this way! User cannot click on number to call



- Dark patterns: deliberate tricks
 - Orchestrated to trick the user
 - Nagging, obstruction, sneaking, **forced action**



3. Open-Source Resources

- Components, frameworks, systems available free of charge
- Community-driven
- Available for interaction design:
 - Design pattern libraries
 - Bootstrap framework

The Practice of Interaction Design

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The Practice of Interaction Design

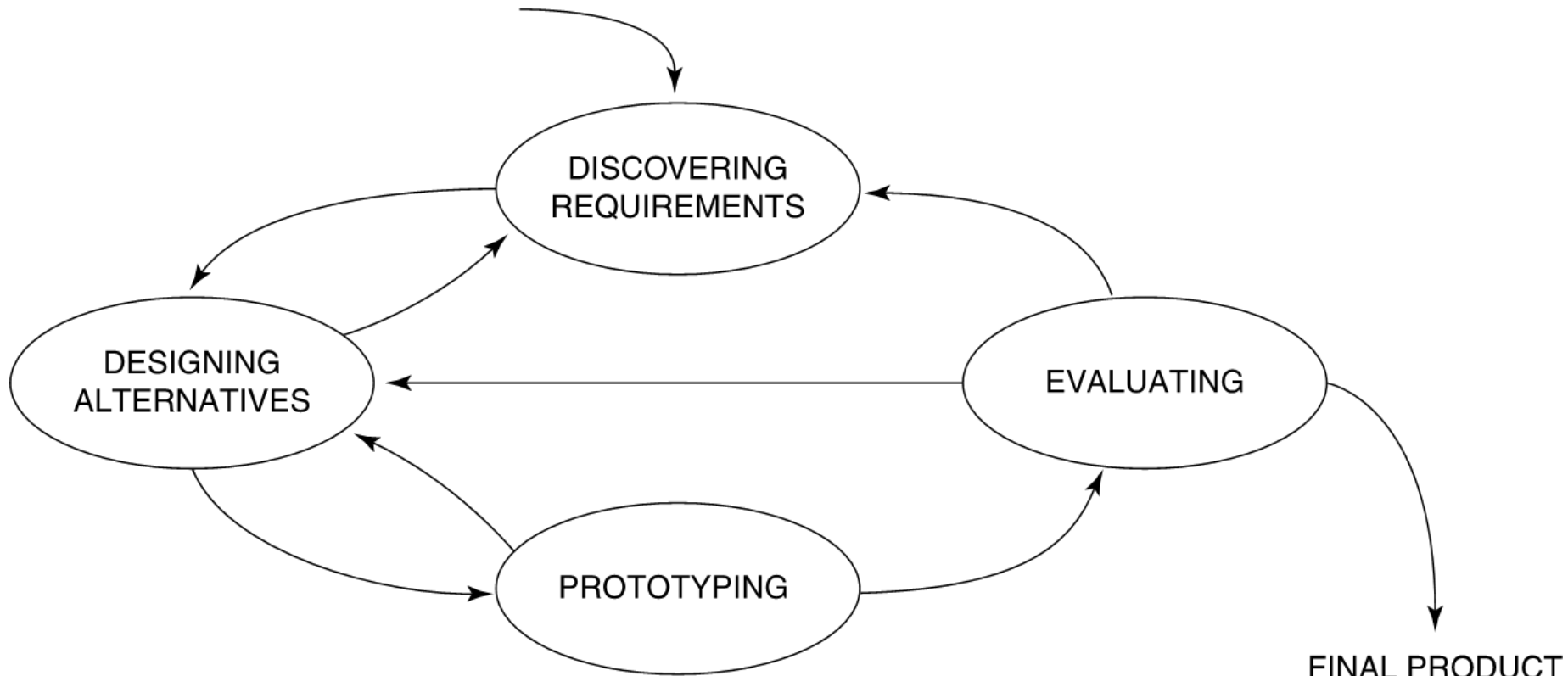
Overview

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- Tools **support** all aspects of the **design process**:
 - Creativity, sketching, simulation, brainstorming, library search, mind mapping, and video capture
- Tools integrate together to **speed up prototyping**
- Tooling landscape changes all the time
- **Interactive wireframes or mockups** can be produced using, for example:
 - Moqup©, Balsamiq©, Axure©
 - Figma
 - Higher-fidelity prototypes can be produced by linking interactive wireframes to design pattern library with software components

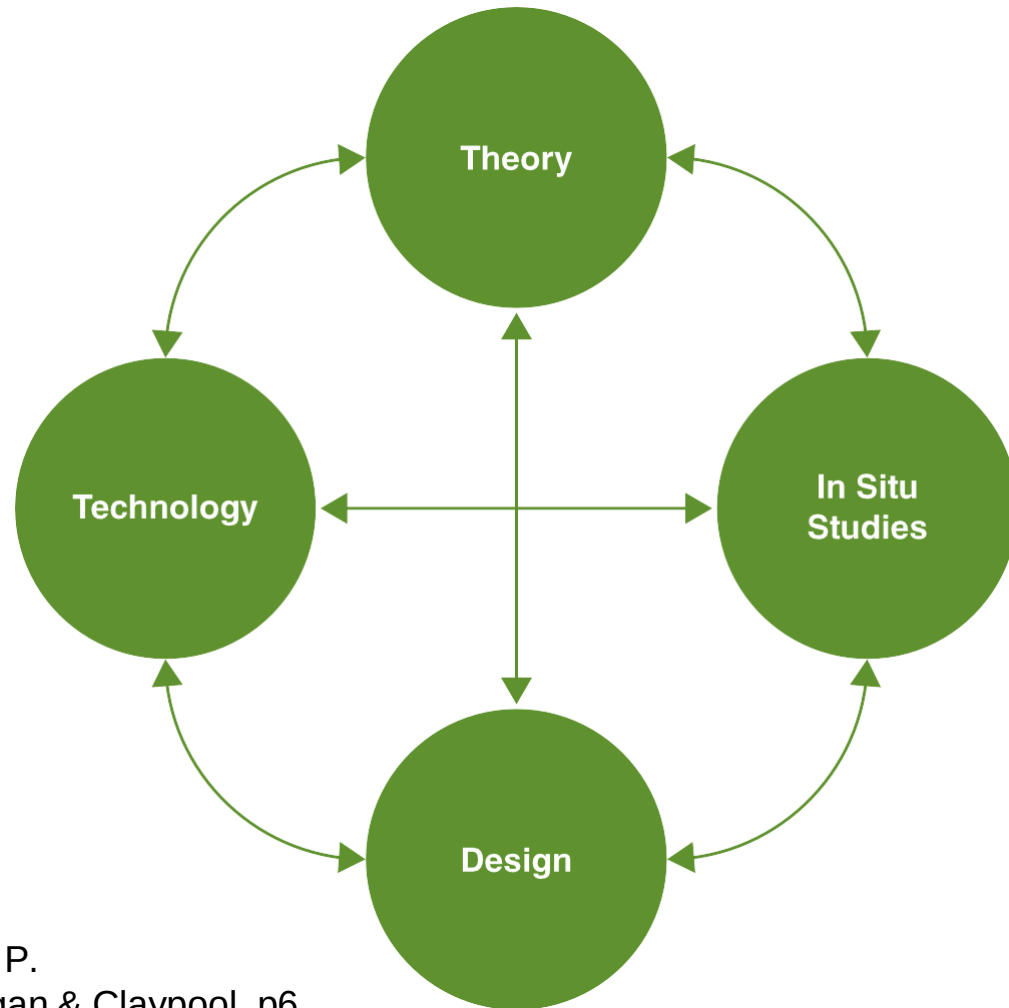
A simple interaction design lifecycle model

Exemplifies a user-centered design approach



Newer lifecycle model: Research in the Wild

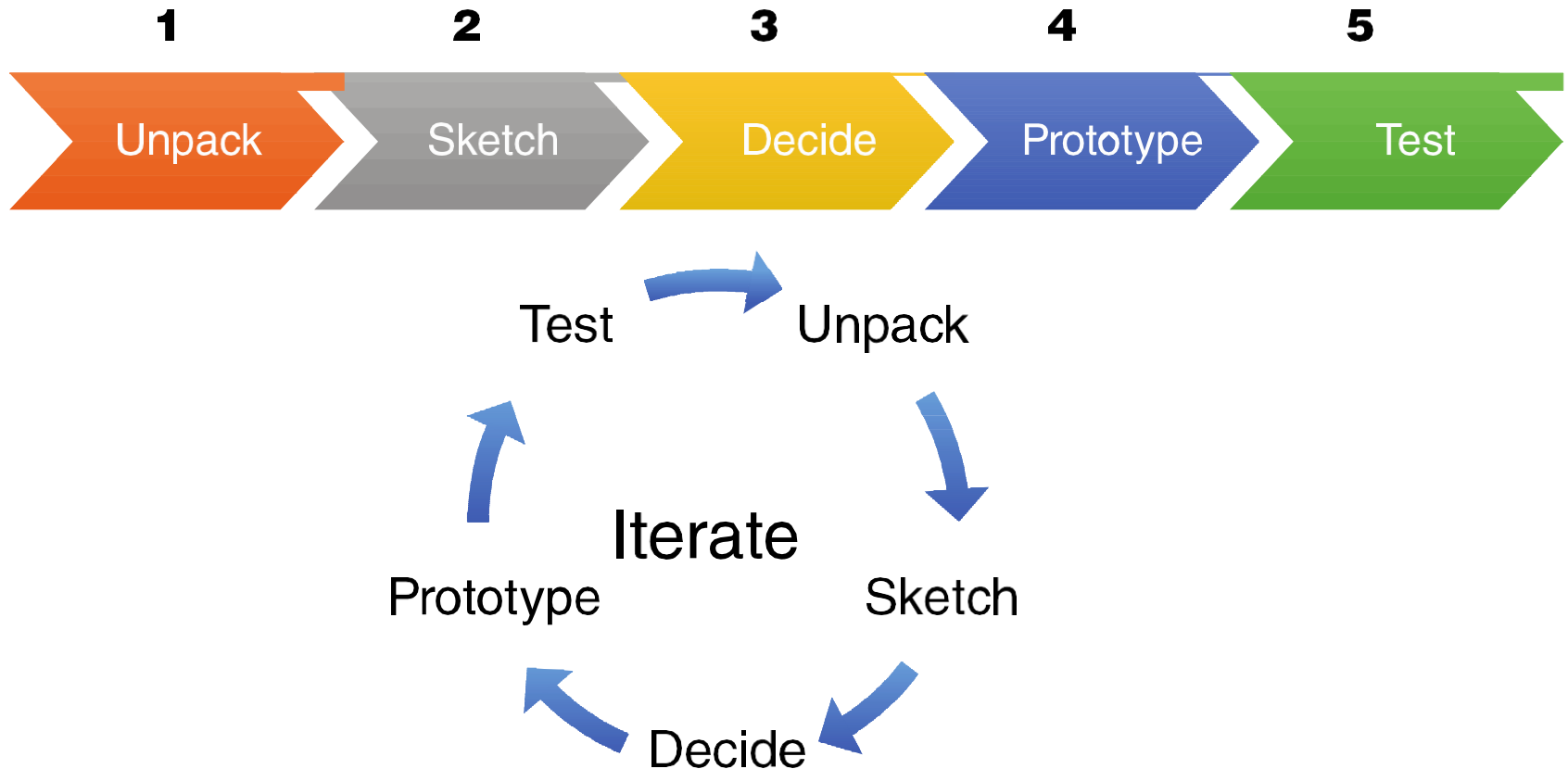
A framework for
research in the
wild studies



Source: Rogers, Y., and Marshall, P.
(2017) *Research in the Wild*. Morgan & Claypool, p6.
(used courtesy of Morgan and Claypool)

Concentrated lifecycle model: Google Design Sprints

Knapp, J. with Zeratsky, J., and Kowitz, B. (2016) *Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days*. Bantam Press, UK.



Source: [Google Design Sprints](#) (used courtesy of Agile Marketing)

Practical issue: Who are the users/stakeholders?

- Not obvious

- **382 distinct types** of users for smartphone apps

Zhao, S., Ramos, J., Tao, J., Jiang, Z., Li, S., Wu, Z., Pan, G., and Dey, A.K. (2016) Discovering Different Kinds of Smartphone Users Through Their Application Usage Behaviors. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '16)*. ACM, New York, NY, pp. 498–509.

- Many products are intended for use by **large** sections of the population, so the user is “**everybody**”
 - More targeted products are associated with specific roles

- Stakeholders

- **Larger** than the group of direct **users**
 - Identifying stakeholders **helps** identify groups to include in interaction design activities

Practical issue: What are the users' needs?

- Users **rarely** know what is possible
- Instead:
 - **Explore** the problem space
 - **Investigate** who are the users
 - Investigate user activities to see what can be improved
 - **Try out ideas** with potential users
- Focus on **peoples' goals, usability, and user experience goals**, rather than expect stakeholders to articulate requirements

Practical issue: How to generate alternatives



- Humans **tend to stick** with something that works
- Considering **alternatives** helps identify better designs
- Where do alternative designs come from?
 - **'Flair and creativity'**: research and synthesis
 - **Cross-fertilization of ideas** from different perspectives
 - **Users** can generate different designs
 - Why not exploit this amazing resource?
 - Product evolution based on changing use
 - **Seek inspiration**: similar products and domain, or different products and domain (what is state of the art?)
- **Balancing** constraints and trade-offs (tough work)

Practical issue: Choosing among alternatives

- Interaction design focuses on **externally-visible** and **measurable** behavior
- Technical feasibility
- Evaluation with users or peers - Prototypes not static documentation (like PID) because behaviour is key
- A/B Testing
 - Online method to inform choice between alternatives
 - Nontrivial to set appropriate metrics and choose user group sets
 - also known as split testing - process of showing two variants of the same web page to different segments of users at the same time and comparing which drives more conversions.
- Quality thresholds
 - Different stakeholder groups have different quality thresholds
 - Usability and user experience goals lead to relevant criteria

Practical issue: How to integrate interaction design activities within other models

- **Integrating** interaction design activities **in lifecycle** models from other disciplines requires **careful planning**
- Software development lifecycle models are prominent
- Integrating with **agile** software development is **promising** because:
 - It incorporates tight iterations
 - It champions early and regular feedback
 - It handles emergent requirements
 - It aims to strike a balance between flexibility and structure
- The whole week of this module will be devoted to topic of ROI (return on investment)

Discovering Requirements

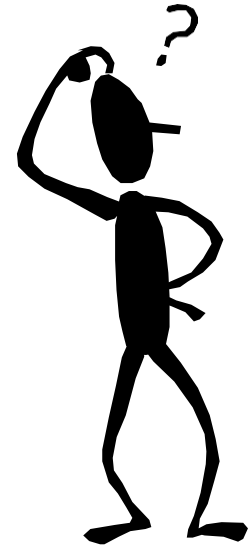
5. What, how and why?

- **Research** shows that, in general, it could generally be **done a lot better**.
- The **importance** of requirements

6. Different types of requirements

7. Data gathering for requirements

- Bringing requirements to life
 - **Personas**
 - **Scenarios**

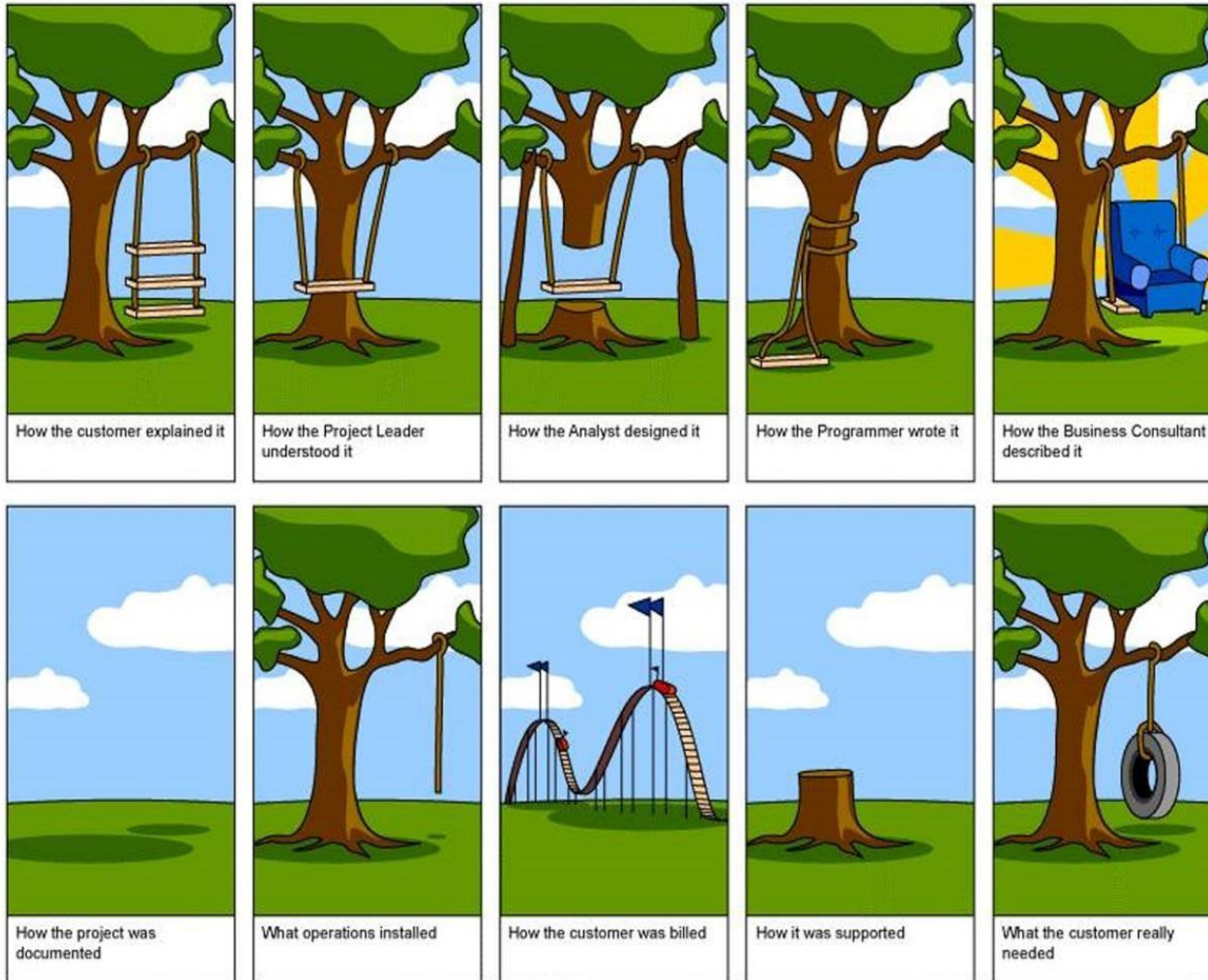


5. What, how and why?

- What is the **purpose** of the **requirements** activity?
 - Explore the problem space
 - Establish a description of what will be developed
- How to capture requirements once discovered
- In prototypes or operational product
- Through structured or rigorous notations
- Different capturing mechanisms emphasize and de-emphasize different aspects

Why bother?

Requirements activity is the stage where miscommunication occurs most commonly



What are requirements?

- A statement about an intended product that specifies what it is expected to do or how it will perform
- Different forms and different levels of abstraction
- User stories (most prevalent in agile development contexts)

- Format:

As a <role>, I want <behavior> so that <benefit>

- Example user stories for a travel organizer might be:

As a <traveler>, I want <to save my favorite airline for all my flights> so that <I will be able to collect air miles>

As a <travel agent>, I want <my special discount rates to be displayed to me> so that <I can offer my clients competitive rates>

Volere shell

Requirement #: 75

Requirement Type: 9

Event/use case #: 6

Description: The product shall issue an alert if a weather station fails to transmit readings.

Rationale: Failure to transmit readings might indicate that the weather station is faulty and needs maintenance, and that the data used to predict freezing roads may be incomplete.

Source: Road Engineers

Fit Criterion: For each weather station the product shall communicate to the user when the recorded number of each type of reading per hour is not within the manufacturer's specified range of the expected number of readings per hour.

Customer Satisfaction: 3

Customer Dissatisfaction: 5

Dependencies: None

Conflicts: None

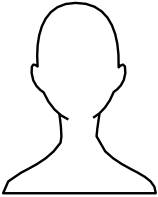
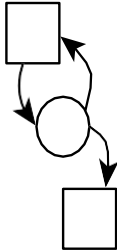
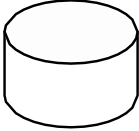
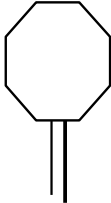
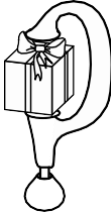
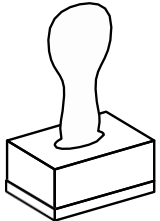
Supporting Materials: Specification of Rosa Weather Station

History: Raised by GBS, 28 July

Volere

Copyright © Atlantic Systems Guild

The seven product dimensions

						
User	Interface	Action	Data	Control	Environment	Quality Attribute
Users interact with the product	The product connects to users, systems, and devices	The product provides capabilities for users	The product includes a repository of data and useful information	The product enforces constraints	The product conforms to physical properties and technology platforms	The product has certain properties that qualify its operation and development

Source: Gottesdiener and Gorman (2012), p.58. Used courtesy of Ellen Gottesdiener

6. Different kinds of requirements

- Functional:
 - What the **system** should **do**
- Data:
 - What kinds of **data** need to be **stored**?
 - **How** will they be stored (for example, database)?
- Environment or context of use:
 - **Physical**: dusty? noisy? vibration? light? heat? humidity? (for example, in a hospital)
 - **Social**: collaboration and co-ordination, data sharing, distributed, synchronous or asynchronous, privacy
 - **Organizational**: user support, communications structure and infrastructure, availability of training
 - **Technical**: On what technologies will it run or need to be compatible?

Different kinds of USER requirements

😊 Users 😊

- Who are they?
- **Characteristics**: nationality, educational background, attitude to computers
- **System use**: novice, expert, casual, frequent
 - Novice**: prompted, constrained, clear
 - Expert**: flexibility, access/power
 - Frequent**: shortcuts
 - Casual/infrequent**: clear menu paths

Different kinds of USER requirements

User profile

- Usability **goals**
- **User experience goals**
- Different products have different requirements and may be implemented in different ways

Usable security

- How to make security **robust** without **detracting** from **user experience**
 - If the usability of security is **ignored**, then security mechanisms will be **circumvented**
 - How many of you try to 'copy and paste' your email address when its requested twice?
 - I do! And I usually get away with it too 😊
 - **Passwords** as an example
 - **Too much advice** about how to choose a password
 - Coping strategies may compromise security

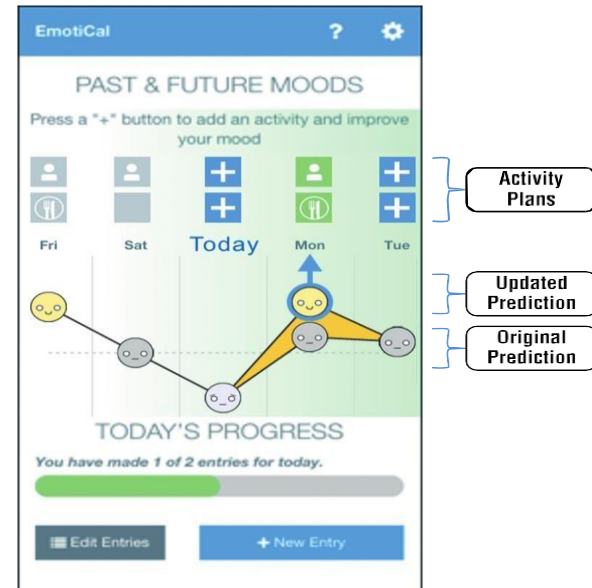
7. Data gathering for requirements

- Interviews, observation, and questionnaires
 - These are **fundamental** means of **requirements gathering**
- Studying documentation:
 - **Procedures** and **rules** are often written down in manuals
 - **Good source of data** about the steps involved in an activity and any regulations governing a task
 - Not to be used in **isolation**
 - Good for understanding legislation and getting **background** information
 - No stakeholder time, which is a limiting factor for other techniques
- Researching similar products:
 - Good for **prompting requirements**

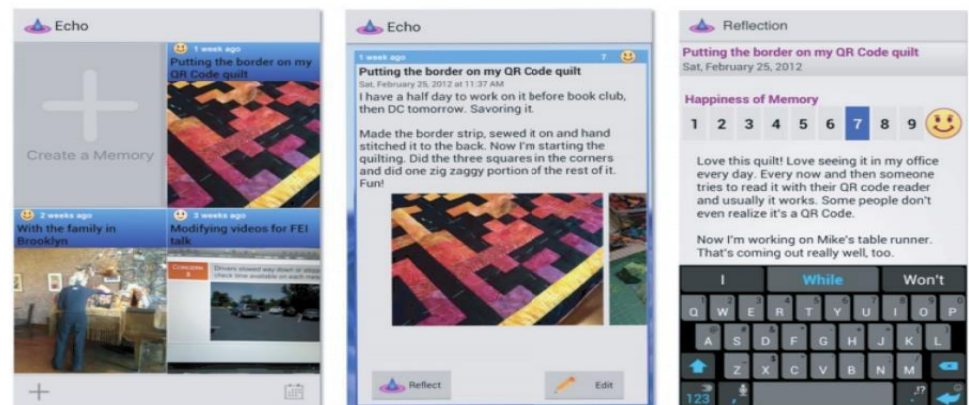
Combining data gathering

The **combination** of different methods is **highly productive**

- direct observation
- indirect observation
- interviews
- diaries
- surveys



(a)



(b)

Source: Hollis et al (2017), Figure 1.
Used courtesy of [Taylor and Francis](#)

Combining data gathering

- **Diaries and interviews:** multiple information devices
- **Interviews, think aloud evaluation, questionnaire, evaluation of working prototype:** memory aid for traumatic brain injury
- **Studying documentation, evaluating other systems, user observation, and group interviews:** ship's maneuvering system
- **Ethnographic study, interviews, usability tests, and user participation:** tabletop user interface for genomic data

Using **probes** to engage with users

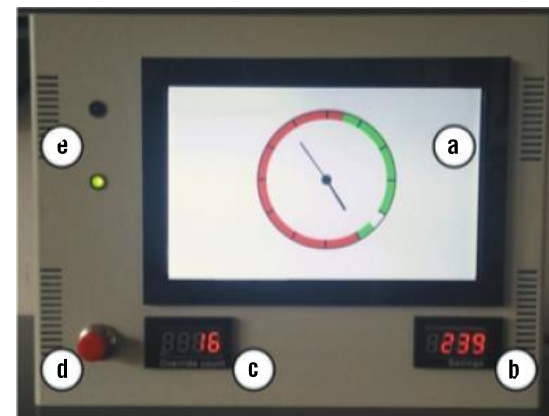
- Many types of probe:
 - Designed to **prompt users** into action
 - For **researchers to learn** about users
- Cultural probe:
 - Wallet containing postcards, maps, camera, photo album, and diary
 - **Participants asked to answer questions using wallet contents**
- Design probe:
 - Form relates specifically to particular question and context, for example, Top Trumps probe



Source: Wallace et al. (2013) Figure 6. Reproduced with permission of [ACM Publications](#).

Using probes to engage with users

- Technology probe:
 - Toolkits, mobile phone apps, sensor-based monitoring, for example, M-Kulinda to alert participants about unexpected movement at home.
- Provocative probe:
 - Technology probe designed to challenge norms and attitudes, for example, the Box to challenge domestic laundry practices



a) electricity status – 12 hour forecast,
b) savings account, c) override button presses,
d) override button, and e) electricity status at the moment

Source: Raptis et al. (2017). Reproduced with permission of [ACM Publications](#).

Contextual Inquiry

- Part of Contextual Design, but also used on its own to gather requirements
- One-on-one field interviews (contextual interviews)
 - 1.5 to 2 hours long
 - Focus on daily life at home or work relevant to the project
 - Uses a model of master (participant) and apprentice (researcher)
- Four main principles:
 - Context: Going to the user, wherever they are, and seeing what they do as they do it
 - Partnership: User and interviewer explore user's life together
 - Interpretation: Observations interpreted by user and interviewer together
 - Focus: Project focus to understand to what should be paid attention

Contextual Inquiry

- Interview guided by “cool concepts” divided into two groups
- Joy of life concepts:
 - How products make our lives richer and more fulfilling
 - Accomplish, connection, identity, and sensation
- Joy of use concepts:
 - Describe impact of using the product
 - Direct in action, the hassle factor, and the learning delta
- Interview in four parts
 - Overview, transition, main interview, and wrap-up
- Following interview, interpretation session
 - Contextual design models are created or consolidated
 - Most relevant models are chosen by team, out of 10 suggested

Brainstorming for innovation

- Include participants from a wide range of disciplines, with a broad range of experience
- Don't ban silly stuff
- Use catalysts for further inspiration
- Keep records. Capture every idea, without censoring
- Sharpen the focus
- Use warm-up exercises and make the session fun

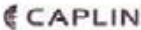
Bringing requirements to life

- Augmenting the basic requirements expressed as stories, in Volere template, or in other form
- Personas
 - Rich descriptions of typical users, not specific people
- Scenarios
 - An informal narrative story, simple, 'natural', personal, and not generalizable

Personas

- Capture a set of user characteristics (user profile)
- Synthesized from real people based on user research
- Typical, not idealized
- Bring to life with name, characteristics, goals, and personal background
 - Relevant to product under development
- Good persona helps designer with design decisions and reminds team about who will use the product
- Develop a small set of personas with one primary

Example Persona #1



Ginnie

Receives private tutoring in Maths and English as these are not her strong subjects. Enjoys playing for the school's 2nd teams for netball and Lacrosse and is good at art.

She loves recording her favourite shows: ER and Sun Valley High on Sky+ and spends some of her time on her Laptop that Daddy bought her watching videos on YouTube, downloading music, keeping up to date with her friends on Facebook and chatting via MS IM to her cousin who is at University in Leeds.

She loves Ugg boots and Abercrombie & Fitch and uses the Internet to shop and find the cheapest prices.

"I want to easily hook up with my friends whilst watching TV"

BACKGROUND

- 15, Female
- Ongoing Private Education
- Ambitious
- Comfortable using technology to communicate

MOTIVATIONS

- Keeping in touch with her network
- Fashion/street cred
- Keeping up with peers.

FRUSTRATIONS

- Sad people trying to be 'friends' on Facebook
- Having to be in bed @ 11pm
- Being swamped in friends updates
- Missing important status updates



Slide 48

Example Persona #2

Family traveler



Organised

Practical

Expects high standard

Goals

- To book comprehensive travel quickly
- To find a trip that meets the needs of the whole family
- To feel supported and guided from the beginning of the booking experience right to the end.

Frustrations

- Wasting time filling in forms
- Too much irrelevant information
- Existing systems tend to be too diverse and complicated

Bio

Will loves to take his family on adventure holidays to explore new challenges. His children, Sky (8) and Eamonn (15) are old enough to take part in several sporting activities and he wants to make the most of this before they no longer want to go on trips with him and his wife, Claire. He likes the fact that choosing travel options is so much easier than it used to be, but is frustrated by the many different sources and disjointed options that this can result in. He wants a travel organiser that can provide clear support for family holidays while offering as wide a choice as possible.

Motivation

Price

Comfort

Choice

Favourite destinations



"I want a travel organiser that will offer me a range of potential vacations that suit our needs"

Age: 35
Work: Plumber
Family: Married, two children

Personality

Introvert	Extrovert
Thinking	Feeling
Sensing	Intuition

Developed using [Xtensio Templates](#)

Please participate by
going to [menti.com](#) –
use the code 3817

Example Scenario for group travel organizer

“The Thomson family enjoy outdoor activities and want to try their hand at sailing this year. There are four family members: Sky (8 years old), Eamonn (12 years old), Claire (32), and Will (35). One evening after dinner they decide to start exploring the possibilities. They want to discuss the options together but Claire has to visit her elderly mother so will be joining the conversation from her mother’s house down the road. As a starting point, Will enters an idea they had been discussing over dinner – a sailing trip for four novices in the Mediterranean. The system supports users to log on from different locations and use different devices so that all members of the family can interact easily and comfortably with it wherever they are. The system's initial suggestion is a flotilla, where several crews (with various levels of experience) sail together on separate boats. Sky and Eamonn aren't very happy at the idea of going on vacation with a group of other people, even though the Thomson’s would have their own boat. The travel organizer shows them descriptions of flotillas from other children their ages and they are all very positive, so eventually, everyone agrees to explore flotilla opportunities. Will confirms this recommendation and asks for detailed options. As it's getting late, he asks for the details to be saved so everyone can consider them tomorrow. The travel organizer emails them a summary of the different options available.”

Scenarios

- May be textual descriptions, animations, audio or video
- Example animation scenarios



Source: Keirnan et al. (2015), Figure 1.
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Scenarios and personas

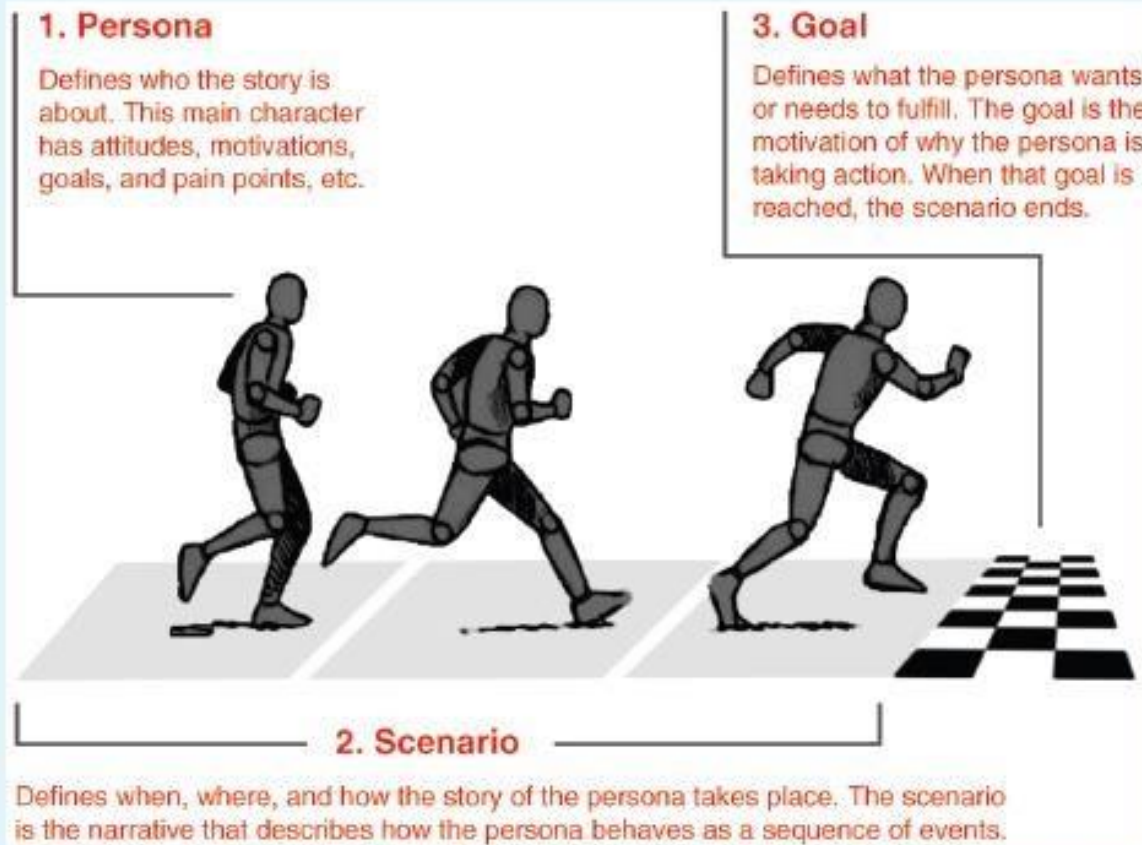


Figure 10.10 The relationship between a scenario and its associated persona

Source: <http://www.smashingmagazine.com/2014/08/06/a-closer-look-at-personas-part-1/>

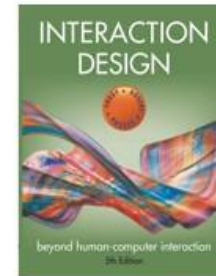
Topics covered this lecture...(Lecture 6)

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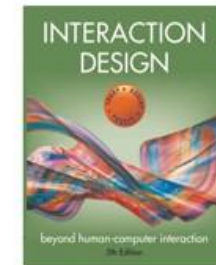
- **Discovering requirements**

- A requirement is a **statement** about an **intended product** that specifies **what it is expected to do** or **how it will perform**.
- Articulating requirements, Different kinds of requirements, Requirements data gathering, Scenarios, Personas.
- Scenarios and personas together **bring requirements to life**.
- **Use cases** capture details.



Chapter 13

INTERACTION
DESIGN
IN PRACTICE



Chapter 11

DISCOVERING
REQUIREMENTS

Any question?

Thank you